

Yue CAO

+33 7 75 79 78 10 | yuec1963@gmail.com | 2063mmz | [in](#) yue-cao | [Paris, France](#)

EXPERIENCE

Berger-Levrault

Feb. 2026 – Aug. 2026

NLP Research Intern

Boulogne, France

- **Project:** Intelligent Assistance Pack Generator for Offline Industrial Maintenance
- Prototyping a system that automatically generates compact assistance packs for offline tiny LLMs deployed on mobile devices, aimed at supporting maintenance technicians in disconnected environments under privacy, latency, and resource constraints
- Analyzing work orders to identify documentary needs and selecting the most relevant content from heterogeneous sources, including technical PDFs, documents, images, and intervention history
- Building, maintaining, and updating offline packs by preparing the chunking and generating the embeddings, indexes, and metadata required by the existing offline engine in its standard format, while taking into account tiny-LLM context constraints
- Optimizing pack quality through compression, deduplication, prioritization, and simple quality indicators to retain only the most useful content within strict size constraints
- Building and testing a RAG pipeline for the generated packs, and integrating it with the ONNX-based deployment flow of the existing Android offline assistant to assess relevance, coherence, response time, and mobile behavior

EDUCATION

INALCO – PluriTAL

Sept. 2025 – Present

Master's (M2) in Natural Language Processing

Paris, France

- **Coursework:** Machine Learning, Neural Networks, Language Modeling, Algorithmic Programming, Object-Oriented Programming, Web Technologies, Linguistics for Multilingual NLP

Université Paris Nanterre – PluriTAL

Sept. 2024 – June 2025

Master's (M1) in Natural Language Processing

Nanterre, France

- **Coursework:** Corpus Processing, Textual Statistics, Mathematics, Parallel Corpora, Treebanks and Parsing, Text Mining, Structured Documents, SQL

Heilongjiang International University

Sept. 2020 – June 2024

Bachelor of Arts in French Language and Literature

Heilongjiang, China

- **Dissertation:** Analyse des méthodes de traduction de métaphore dans les langues familières à l'exemple du feuillet français *Dix pour Cent*

PROJECTS

RAG Pipeline for Question Answering

2026

- Built a complete RAG pipeline using ChromaDB and Gemini for question answering over the novel *My Brilliant Friend*
- Chunking, embedding, vector indexing, and prompt orchestration for grounded answers

Transformer Models for Classical Chinese

2026

- Trained BERT- and GPT-style Transformers from scratch for genre classification and classical Chinese text generation; experimented on historical prose and Tang poetry

Image Classification on CIFAR-10 with CNN

2026

- Trained and evaluated a CNN on CIFAR-10 using PyTorch, and improved performance through architecture tuning and training refinements

- Applied data augmentation, shuffled mini-batches, and ReLU-based optimization strategies to improve classification results

Urban Traffic Forecasting in Paris Using Traffic and Weather Time Series

2026

- Built a short-term urban traffic forecasting pipeline using Paris Open Data traffic counts and Open-Meteo weather variables
- Preprocessed raw JSON traffic data, aligned traffic and weather time series, and engineered temporal and seasonal features with cyclic encoding
- Trained and compared a historical baseline, HistGradientBoosting, LSTM, and GRU models for multivariate time-series forecasting
- Developed a Streamlit application to load trained models and generate predictions without retraining

SKILLS

Programming : Python (pandas, NumPy, scikit-learn, Keras, NLTK, spaCy, Stanza, Word2Vec, Transformers), SQL, Bash/Shell, HTML, CSS, XML

Deep Learning : CNN, RNN, LSTM/GRU, Transformer architectures (BERT, GPT)

NLP Tasks : RAG, sentiment classification, topic modeling, text generation, information extraction, OCR, ASR, TTS, speech synthesis

Tools : VS Code, Jupyter, Overleaf, Hugging Face, Oxygen XML, MySQL, Praat, Tesseract, Unitex, marimo, neo4j, Nooj

Languages : Chinese (native), French (Professional), English (Professional)